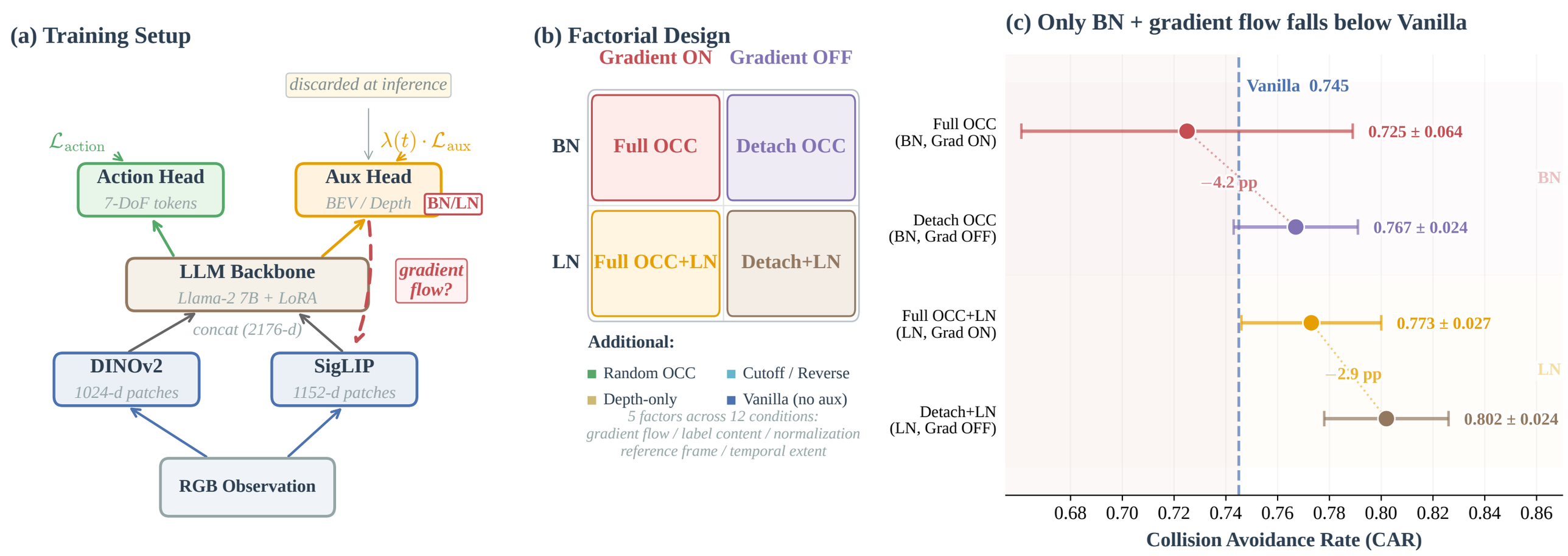

(a) Training Setup
discarded at inference
$\mathcal{L}_{\text{action}}$
$\lambda(t) \cdot \mathcal{L}_{\text{aux}}$
Action Head
7-DoF tokens
Aux Head
BEV / Depth
BN/LN
LLM Backbone
Llama-2 7B + LoRA
gradient flow?
concat (2176-d)
DINOv2
1024-d patches
SigLIP
1152-d patches
RGB Observation
(b) Factorial Design
Gradient ON
Gradient OFF
BN
LN
Full OCC
Detach OCC
Full OCC+LN
Detach+LN
Additional:
Random OCC
Cutoff / Reverse
Depth-only
Vanilla (no aux)
5 factors across 12 conditions:
gradient flow / label content / normalization
reference frame / temporal extent
(c) Only BN + gradient flow falls below Vanilla
Vanilla 0.745
Full OCC
(BN, Grad ON)
0.725 ± 0.064
−4.2 pp
BN
Detach OCC
(BN, Grad OFF)
0.767 ± 0.024
Full OCC+LN
(LN, Grad ON)
0.773 ± 0.027
−2.9 pp
LN
Detach+LN
(LN, Grad OFF)
0.802 ± 0.024
0.68
0.70
0.72
0.74
0.76
0.78
0.80
0.82
0.84
0.86
Collision Avoidance Rate (CAR)